Organisation of the organism

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Cell structure [1]

Which structure is found in a typical plant cell but is **not** found in an animal cell?

(A) cell membrane

(B) cell wall

(C) mitochondrion

(D) ribosome

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2 · Part 1 Core Cell structure [1]

Which row shows the structures in a bacterial cell?

	cell wall	nucleus	plasmids
A	present	present	absent
B	absent	absent	present
C	absent	present	present
D	present	absent	present

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3 · Part 1 Core Cell structure [1]

What is the function of ribosomes?

(A) aerobic respiration

(B) controlling which substances enter and leave the cell

(C) protein synthesis

(D) storing cell sap

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4 · Part 1 Core Cell structure [1]

Cells in the muscle of a bird's wing contain many more mitochondria than cells in its fat-storage tissue.

What is the reason for this?

A The muscle cells carry out more aerobic respiration to release energy for contraction.

B The muscle cells make more protein.

C The muscle cells contain more genetic material.

D The muscle cells need to control which substances enter them more closely.

5 · Part 1 Core Cell structure [1]

Which specialised cell is matched to its correct function?

A neurone – transport of oxygen

B root hair cell – photosynthesis

C red blood cell – conduction of electrical impulses

D ciliated cell – movement of mucus in the trachea and bronchi

6 · Part 1 Core Cell structure [1]

Which row gives a correct example of a tissue, an organ and an organ system?

	tissue	organ	organ system
A	stomach	muscle tissue	digestive system
B	muscle tissue	stomach	digestive system
C	muscle tissue	digestive system	stomach
D	digestive system	stomach	muscle tissue

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7 · Part 1 Core Cell structure [1]

Which statement about cells is correct?

A New cells are produced by division of existing cells.

B All cells contain a nucleus.

C Only plant cells contain mitochondria.

D Animal cells have a cell wall made of cellulose.

8 · Part 1 Core Size of specimens [1]

In a drawing, a hair from the surface of a leaf is 36 mm long. The actual length of the hair is 0.45 mm.

What is the magnification of the drawing?

A × 0.0125

B × 16.2

C × 80

D × 800

9 · Part 1 Core Size of specimens [1]

The magnification of a photograph of a pollen grain is × 250. The diameter of the pollen grain in the photograph is 15 mm.

What is the actual diameter of the pollen grain?

A 0.06 mm

B 0.6 mm

C 16.7 mm

D 3750 mm

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10 · Part 1 Extended Size of specimens [1]

The actual length of a chloroplast is 5 µm. In a photograph taken with an electron microscope, the chloroplast is 40 mm long.

What is the magnification of the photograph?

A × 8

B × 80

C × 800

D × 8000

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11 · Part 1 Core Cell structure [5]

A student compares a cell from the leaf of a pondweed with a cell from a human liver.

(a) State **two** structures, other than vacuoles, that are found in both cells. [2]

(b) State **two** structures that are found in the pondweed leaf cell but are **not** found in the liver cell. Do not include vacuoles in your answer. [2]

(c) Describe how the vacuole in the pondweed leaf cell is different from the vacuoles in the liver cell. [1]

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12 · Part 1 Core Cell structure [6]

Bacteria live on the surface of human skin. Bacterial cells are different from the skin cells they live on.

(a) The DNA in a bacterial cell is not inside a nucleus. Describe the forms in which DNA is found in a bacterial cell. [2]

(b) State **two** structures that are found in both a bacterial cell and a human skin cell. [2]

(c) State **one** structure, other than a nucleus, that is found in a human skin cell but not in a bacterial cell. [1]

(d) State **one** structure, other than circular DNA and plasmids, that is found in a bacterial cell but not in a human skin cell. [1]

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13 · Part 1 Core Cell structure [6]

Each structure in a cell has its own function.

(a) Some chemicals damage the cell membrane. Suggest **two** effects on a cell if its cell membrane is damaged. [2]

(b) Most of the chemical reactions in a cell take place in one part of the cell. Name this part of the cell. [1]

(c) Describe the function of the nucleus. [2]

(d) State the function of the large permanent vacuole in a plant cell. [1]

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14 · Part 1 Core Cell structure [7]

The table describes four specialised cells, P, Q, R and S.

cell	description
P	has many tiny hair-like structures on its surface; found lining the trachea
Q	has a cell body with one very long, thin fibre that can reach from the spinal cord to the foot
R	column-shaped and packed with chloroplasts; found in a layer just below the upper surface of a leaf
S	on the outside of a root; has a long, thin extension that grows out between soil particles

(a) Name cells P, Q, R and S. [2]

(b) State the function of cell Q. [1]

(c) Suggest how the long, thin extension of cell S helps the cell to carry out its function. [2]

(d) Sperm cells and egg cells are also specialised cells. State the general name for sperm cells and egg cells, and state their function. [2]

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15 · Part 1 Core Cell structure [7]

The list shows four structures found in the human body.

- stomach
- muscle tissue
- digestive system
- red blood cell

(a) Describe what is meant by the term **tissue**. [2]

(b) Describe what is meant by the term **organ**. [2]

(c) Identify the structure in the list that is an organ, and the structure in the list that is an organ system. [2]

(d) Plants also have organs. Name **one** plant organ. [1]

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16 · Part 1 Core Cell structure [6]

Three different cells, X, Y and Z, were examined. One cell came from a plant, one from an animal and one was a bacterium. The table shows the results.

structure	cell X	cell Y	cell Z
cell wall	present	absent	present
cell membrane	present	present	present
nucleus	present	present	absent
chloroplasts	absent	absent	absent
mitochondria	present	present	absent
ribosomes	present	present	present
plasmids	absent	absent	present
large permanent vacuole	present	absent	absent

(a) Identify which cell is the bacterium. Give **two** reasons from the table for your answer. [3]

(b) Cell X came from a plant, but it has no chloroplasts. Suggest which part of the plant cell X came from, and explain your answer. [2]

(c) Give **one** reason from the table why cell Y must be the animal cell and not the plant cell. [1]

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17 · Part 1 Core Size of specimens [4]

A student makes a drawing of a human egg cell. The diameter of the egg cell in the drawing is 60 mm. The actual diameter of the egg cell is 0.12 mm.

(a) State the formula used to calculate magnification. [1]

(b) Calculate the magnification of the drawing. [2]

(c) The student makes a second, larger drawing of the same egg cell. State what happens to the magnification of the second drawing compared with the first. [1]

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18 · Part 1 Core Size of specimens [6]

The table gives information about photographs of three small animals.

animal	length in photograph / mm	magnification	actual length / mm
flea	50	× 20	to be calculated in (a)
mite	36	to be calculated in (b)	0.4
ant	to be calculated in (c)	× 15	3.2

(a) Calculate the actual length of the flea. [2]

(b) Calculate the magnification of the photograph of the mite. [2]

(c) Calculate the length of the ant in its photograph. [2]

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19 · Part 1 Extended Size of specimens [6]

A scientist uses an electron microscope to photograph part of a liver cell. The magnification of the photograph is × 25 000.

In the photograph, a mitochondrion is 75 mm long and a ribosome has a diameter of 0.5 mm.

(a) Calculate the actual length of the mitochondrion in mm. [2]

(b) Convert your answer to (a) into µm. [1]

(c) Calculate the actual diameter of the ribosome in µm. [2]

(d) Calculate how many times longer the mitochondrion is than the diameter of the ribosome. [1]

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20 · Part 1 Extended Size of specimens [6]

The actual diameter of a red blood cell is 7.5 µm. A student draws the red blood cell with a diameter of 30 mm.

(a) Convert 7.5 µm into mm. [1]

(b) Calculate the magnification of the drawing. [2]

(c) Another student calculated the magnification as × 4. Explain the mistake this student made. [1]

(d) The first student then draws a bacterial cell at the same magnification. The bacterial cell is 6 mm long in the drawing. Calculate the actual length of the bacterial cell in µm. [2]

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